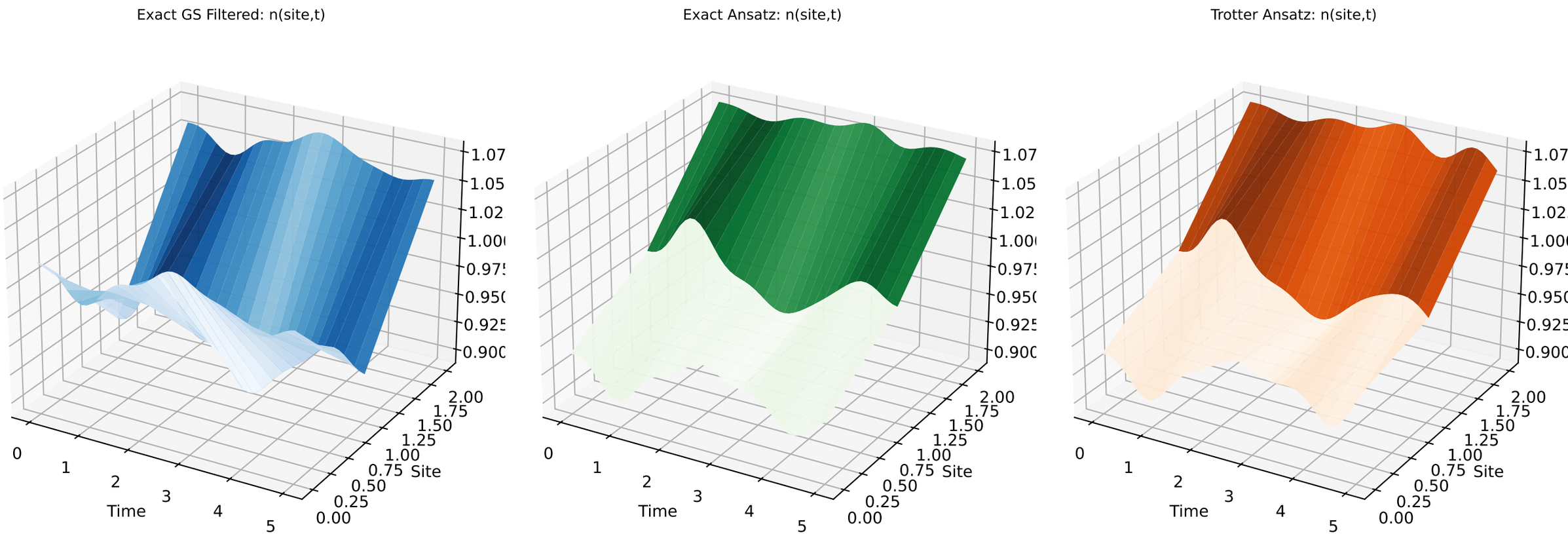
Executed Command

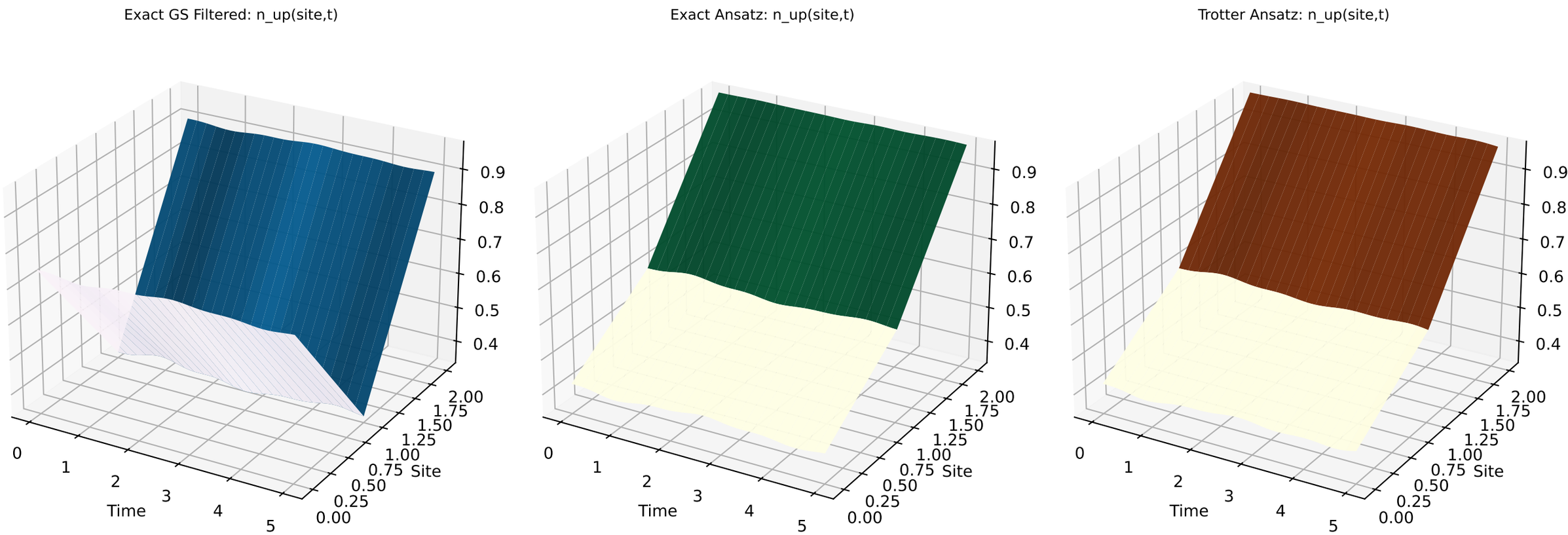
Reference: pipelines/PIPELINE_RUN_GUIDE.md
Script: pipelines/hardcoded_hubbard_pipeline.py

```
/opt/anaconda3/bin/python pipelines/hardcoded_hubbard_pipeline.py --L 3 --t 1.0 --u 4.0 --dv 0.0 --boundary
  periodic --ordering blocked --t-final 5.0 --num-times 51 --suzuki-order 2 --trotter-steps 32 --term-order
  sorted --enable-drive --drive-A 0.2 --drive-omega 1.7 --drive-tbar 4.0 --drive-phi 0.0 --drive-pattern
  staggered --drive-time-sampling midpoint --vqe-reps 2 --vqe-restarts 3 --vqe-maxiter 600 --skip-qpe --initial-
  state-source vqe --fidelity-subspace-energy-tol 1e-8 --output-json
  artifacts/json/H_L3_vt_t1.0_U4.0_S32_reg2.json --output-pdf artifacts/pdf/H_L3_vt_t1.0_U4.0_S32_reg2.pdf
```

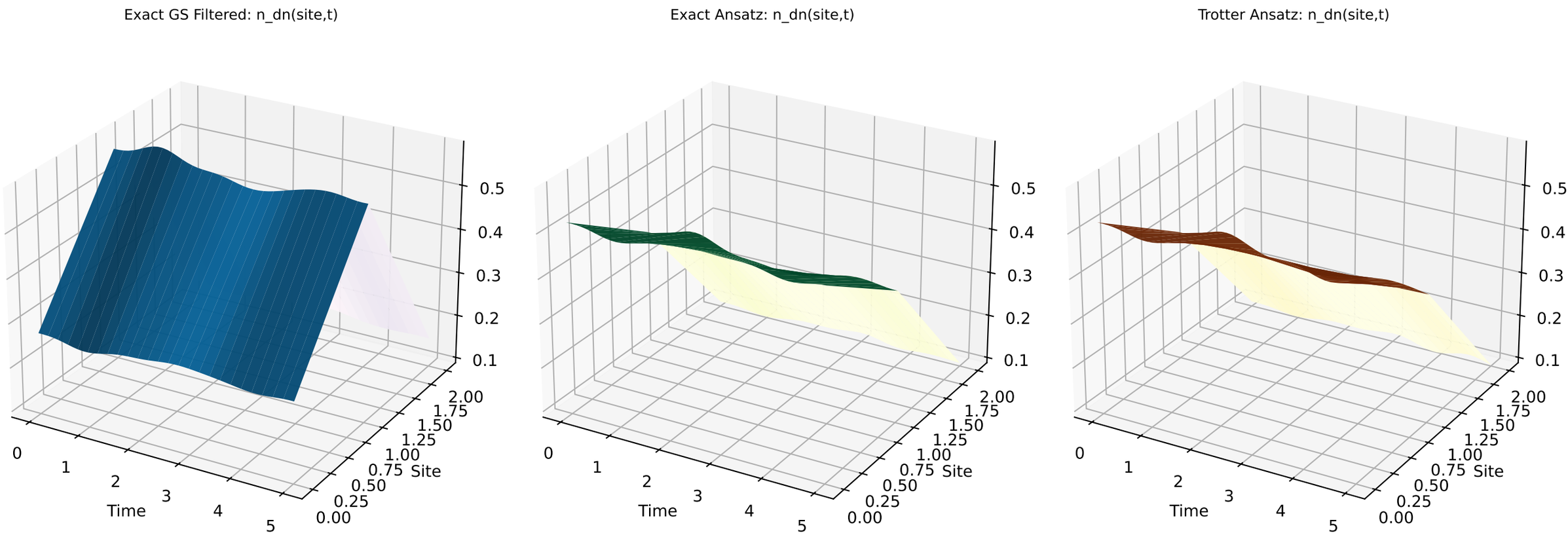
L=3 3D Densities (Total n)



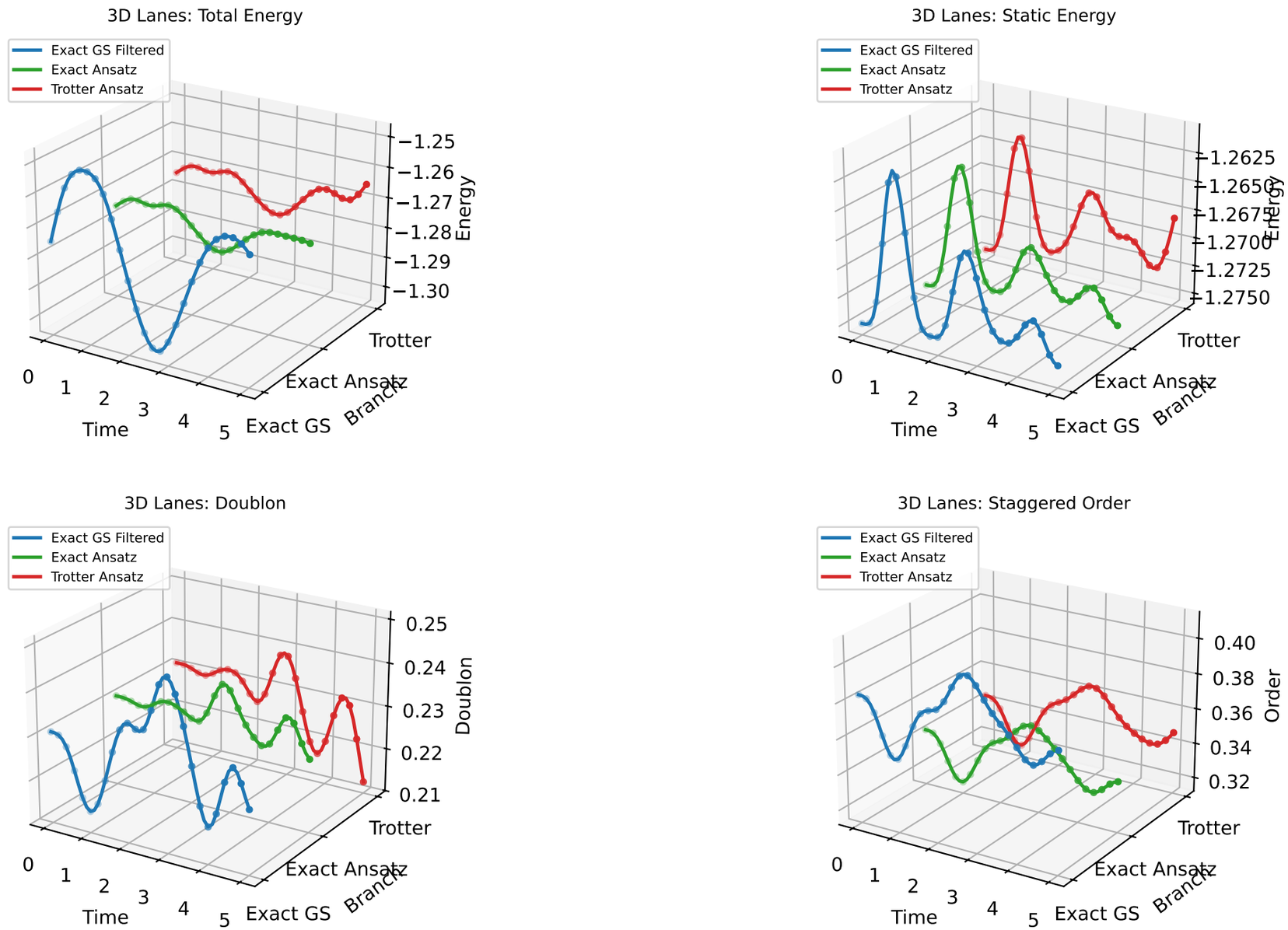
L=3 3D Densities (Spin-Up)

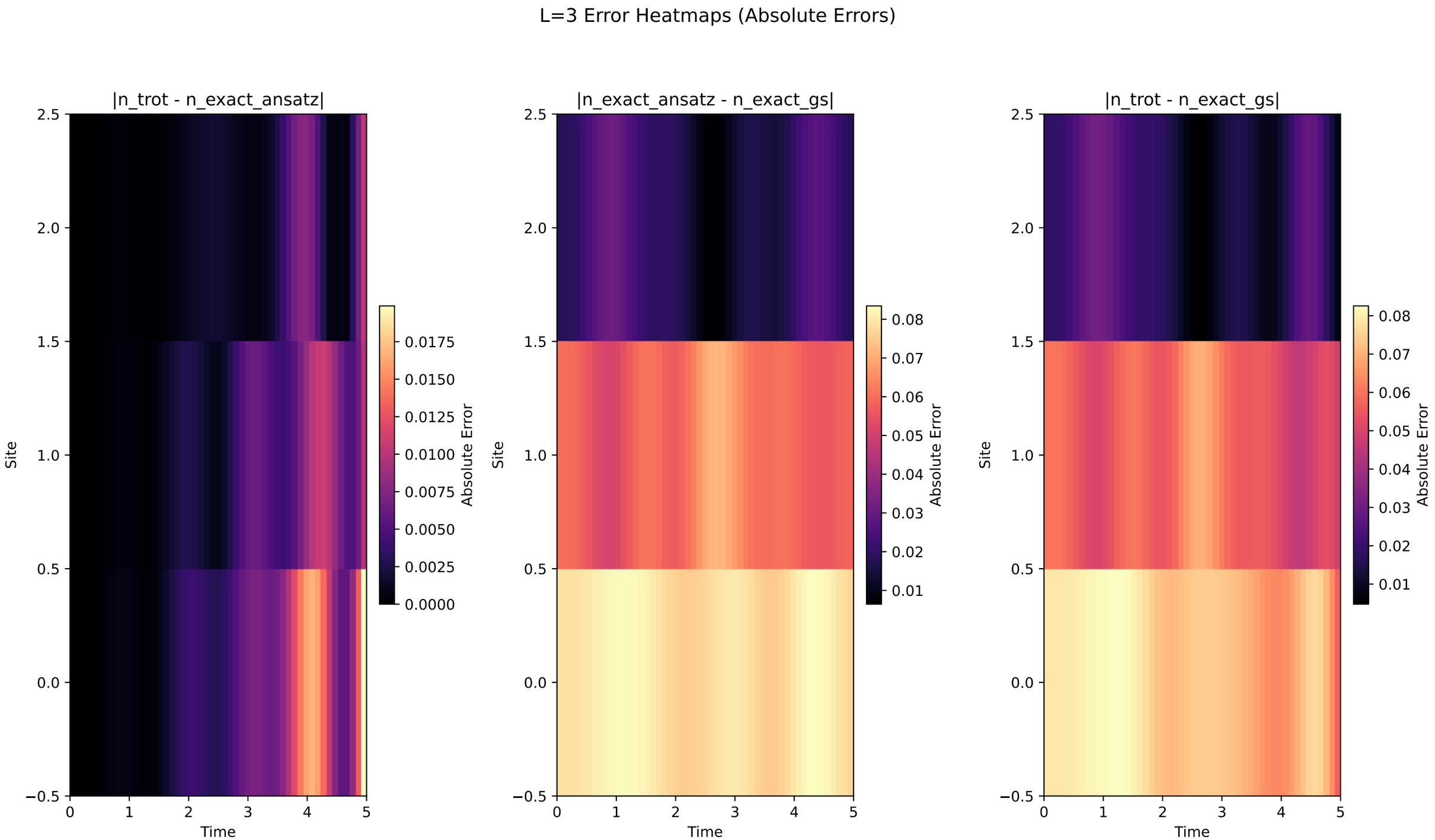


L=3 3D Densities (Spin-Down)

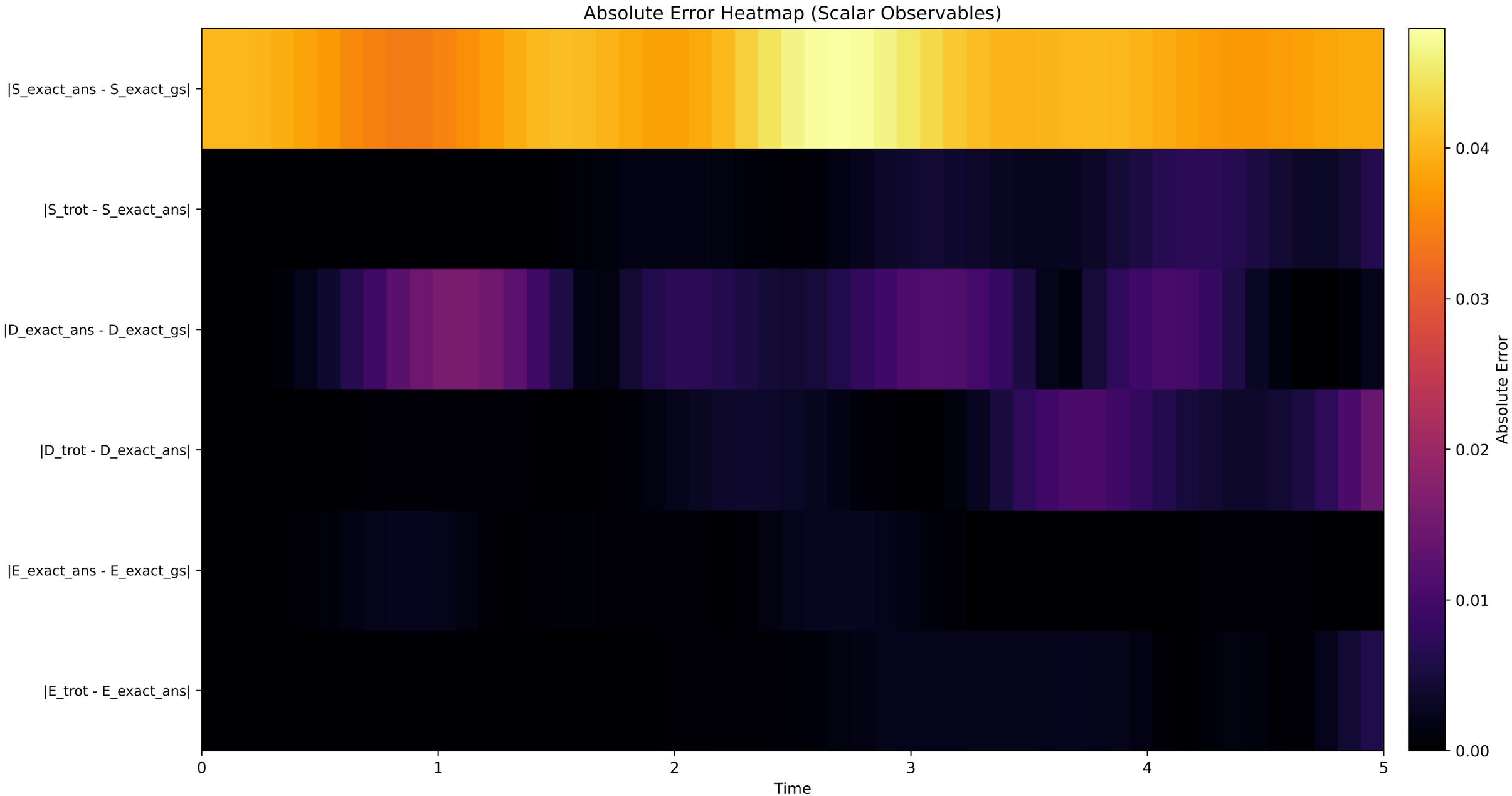


L=3 3D Scalar Observables (Three Evolutions)

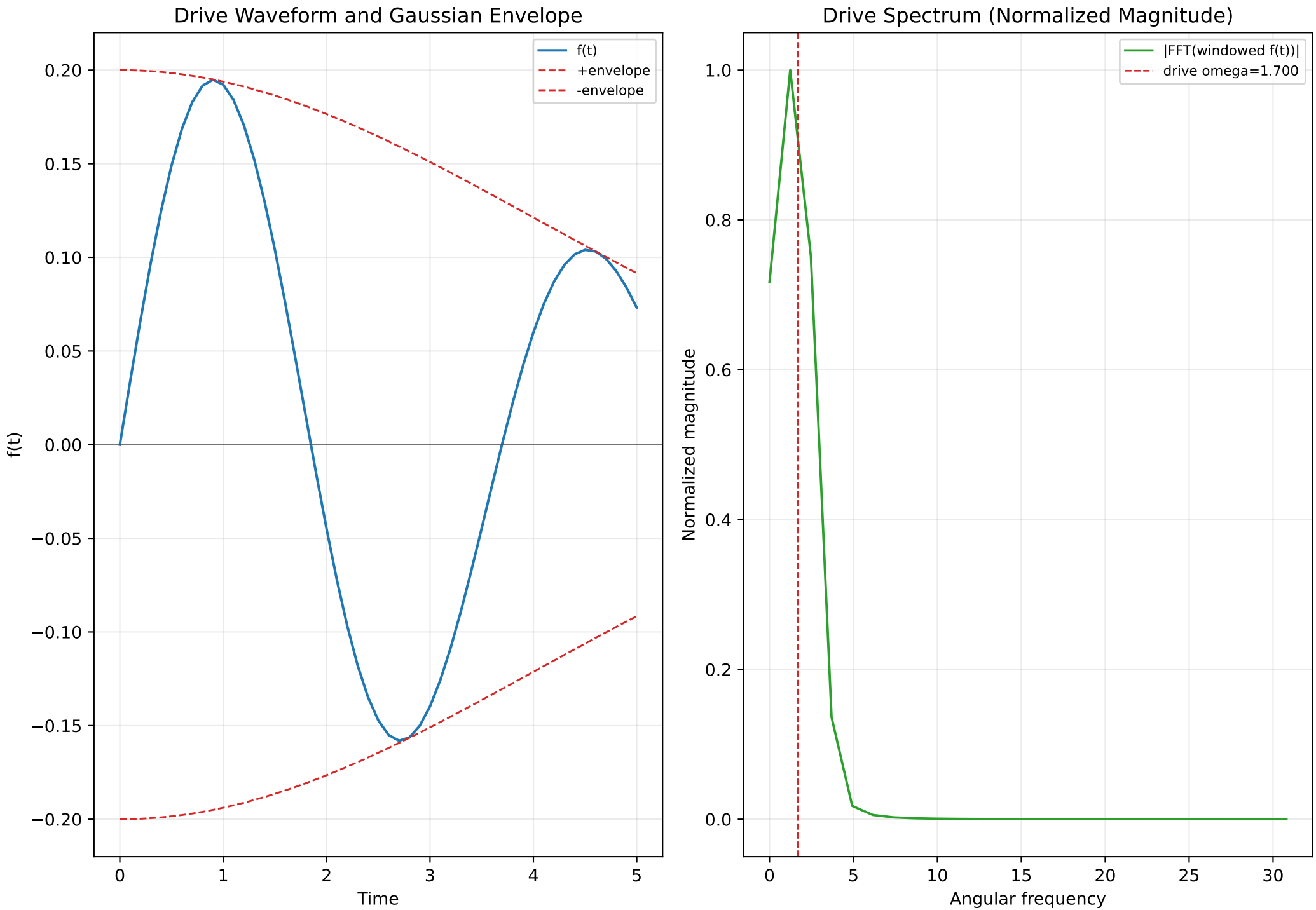


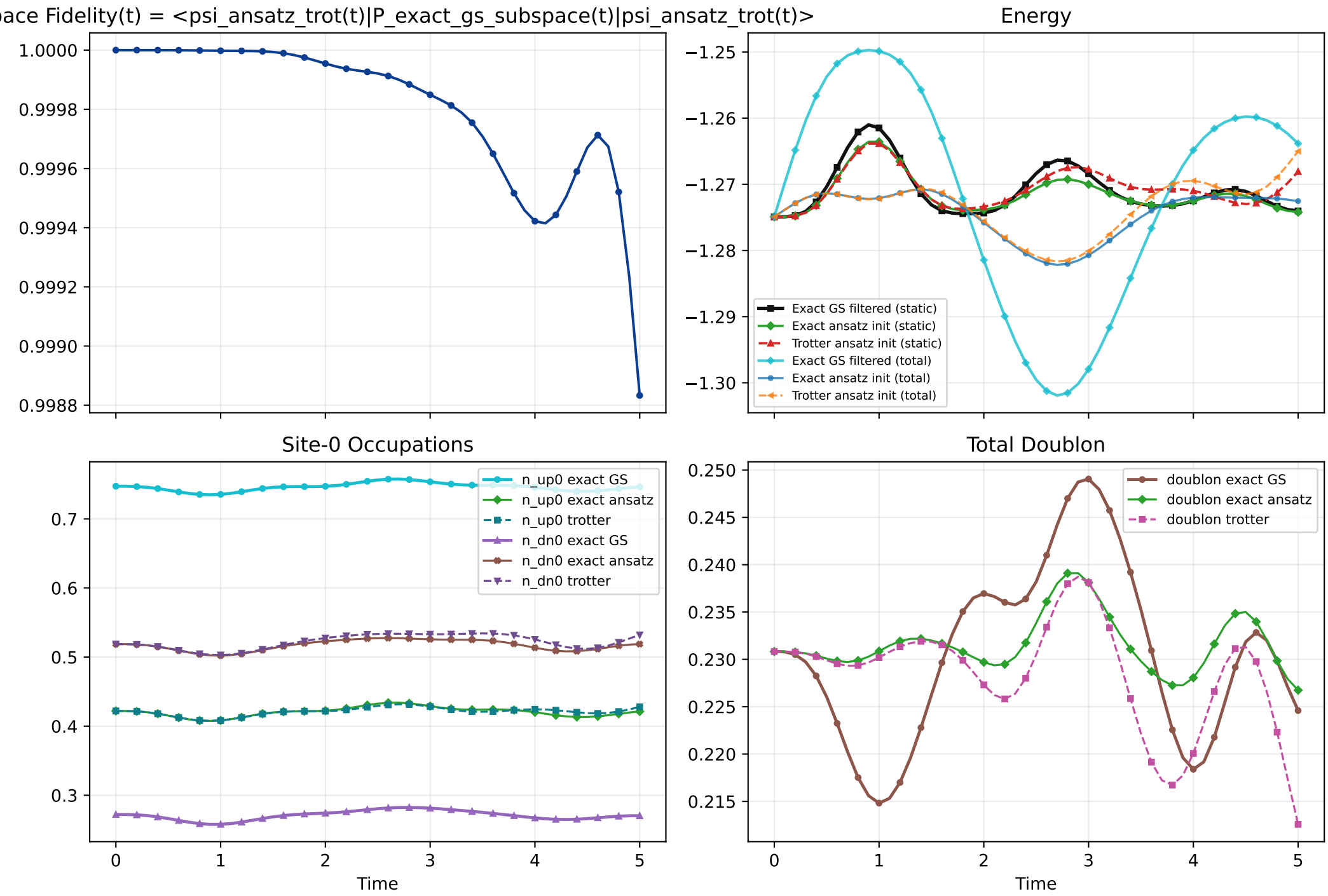

L=3 Error Heatmaps (Absolute Errors)
|n_trot - n_exact_ansatz|
|n_exact_ansatz - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
1
2
3
4
5
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01

L=3 Scalar Error Heatmap



Drive Diagnostics: A=0.2, omega=1.7, tbar=4.0, phi=0.0, t0=0.0

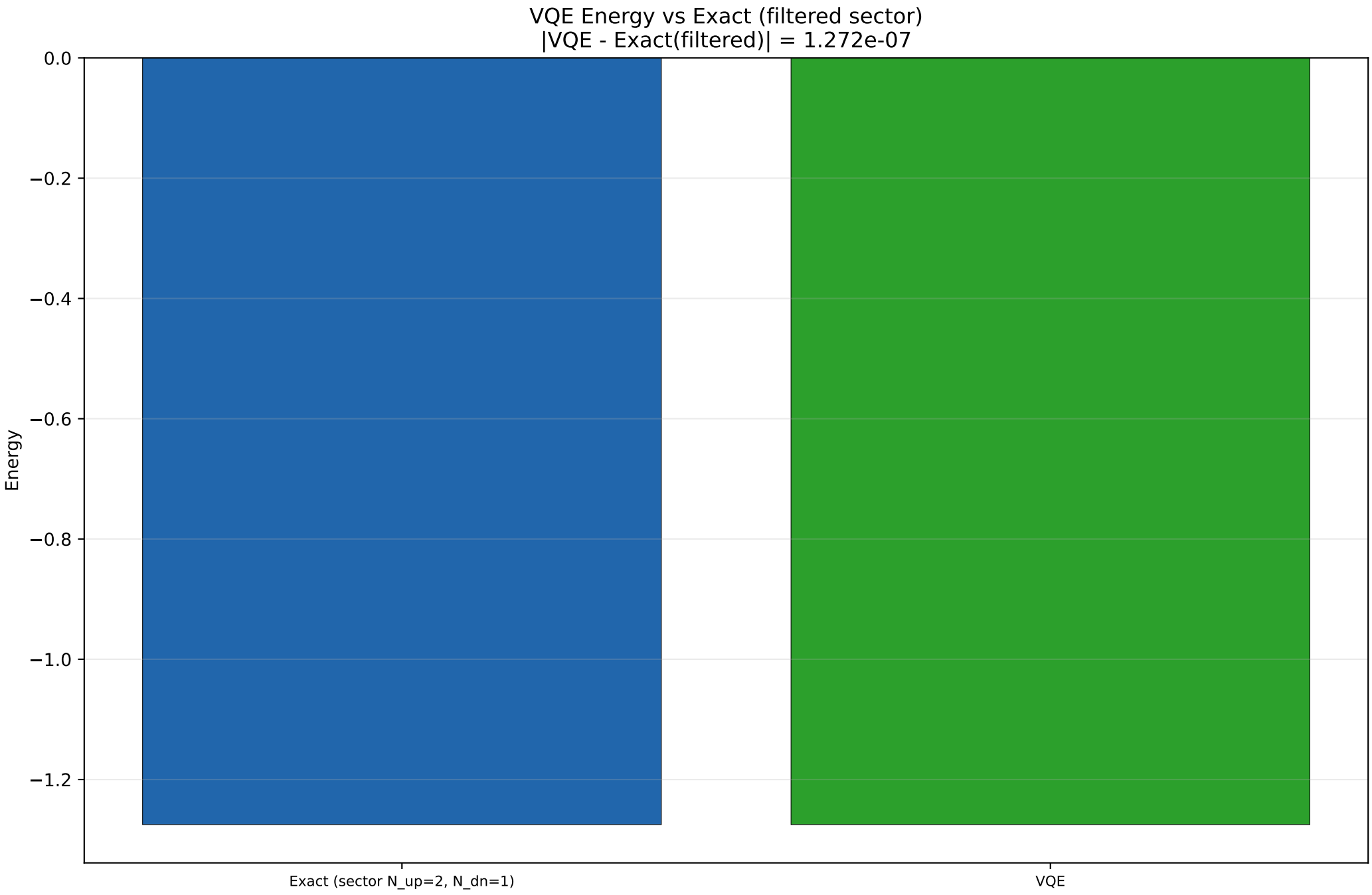



Hardcoded Hubbard Pipeline: L=3
pace Fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>
1.0000
0.9998
0.9996
0.9994
0.9992
0.9990
0.9988
Energy
−1.25
−1.26
−1.27
−1.28
−1.29
−1.30
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
Exact GS filtered (total)
Exact ansatz init (total)
Trotter ansatz init (total)
Site-0 Occupations
n_up0 exact GS
n_up0 exact ansatz
n_up0 trotter
n_dn0 exact GS
n_dn0 exact ansatz
n_dn0 trotter
0.7
0.6
0.5
0.4
0.3
0
1
2
3
4
5
Time
Total Doublon
doublon exact GS
doublon exact ansatz
doublon trotter
0.250
0.245
0.240
0.235
0.230
0.225
0.220
0.215
0
1
2
3
4
5
Time

Appendix — L=3

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

settings: {"L": 3, "t": 1.0, "u": 4.0, "dv": 0.0, "boundary": "periodic", "ordering": "blocked", "t_final": 5.0, "num_times":
exact_trajectory_label: Exact_Hardcode
exact_trajectory_method: python_matrix_eigendecomposition
fidelity_definition: fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>, where P_exact_gs_subspace(t
subspace_fidelity_at_t0: 0.9999999180073835
energy_t0_exact_gs: -1.2749172176353751
energy_t0_exact_ansatz: -1.274917090434822
energy_t0_trotter: -1.2749170904348222
ground_state_exact_energy (full Hilbert): -3.123105625618
ground_state_exact_energy_filtered: -1.2749172176353751
filtered_sector: {'n_up': 2, 'n_dn': 1}
vqe_energy: -1.27491709043483
qpe_energy_estimate: None
initial_state_source: vqe
hamiltonian_terms: 22
reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```